

# Conditional Probability & Independence

CSU, Chico Math 350

2026-02-09

# outline

## Conditional Probability

example 1

example 2

example 3

## Independence

## Example Problems, Pt 1

## Simulation

## Law of Total Probability

## Bayes' Theorem

## Example Problems, Pt 2

## References

# outline

## Conditional Probability

example 1

example 2

example 3

## Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References

# Motivating Conditional Probability

Wouldn't it be great if there was a formal method to update probabilities given new information? Consider this scenario. You draw a card from a deck of 52 cards **and then** learn that the card you drew (still face down) is a face card. What is the probability that this card is a king?

- ▶ There's 12 face cards.
- ▶ 4 of the 12 face cards are kings.
- ▶ Therefore, there's a  $1/3$  probability that the card you drew is a king.

# Motivating Conditional Probability

Without any information that the card you drew is a face card, the probability of drawing a king is  $4/52 = 0.08$ . But if you do for a fact know that the card you have drawn is a face card, the probability of it being a king is  $1/3 = 0.33$ .

This new probability is **conditional** on knowing whether you have drawn a face card.

# Conditional Probability

Conditional probability allows us to update probabilities given a new set of information; e.g. we learned that the card we drew was a face card.

## conditional probability

For any two events  $A$  and  $B$

$$P(B|A) = \frac{P(A \cap B)}{P(A)},$$

so long as  $P(A) > 0$ .

## Conditional Probability, example

Let  $P(A)$  be the probability that we draw a face card and let  $P(B)$  be the probability that we draw a king.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(\text{face card and king})}{P(\text{face card})} = \frac{4/52}{12/52}.$$

## Conditional Probability, example 2

Suppose you are working with a traffic engineer to synchronize two consecutive intersections' stop lights. The planner claims that if the first light is green the second light will be green with probability 0.80. Assume the probability of finding the first light green is 0.60. What is the probability that any given driver will find both lights green?

Let  $P(A) = 0.60$  and  $P(B|A) = 0.80$ . We want to find  $P(A \cap B)$ .



## Conditional Probability, example 3

Sometimes, your data will take the form of contingency tables, and these can be interpreted with regards to conditional probability as well. From the following table, calculate and interpret  $P(\text{tea})$ ,  $P(\text{coffee})$ ,  $P(\text{coffee} \cap \text{tea})$ ,  $P(\text{coffee}|\text{tea})$ .

|                  | coffee | coffee <sup>c</sup> |      |
|------------------|--------|---------------------|------|
| tea              | 150    | 50                  | 200  |
| tea <sup>c</sup> | 650    | 150                 | 800  |
|                  | 800    | 200                 | 1000 |

Table 1: Survey of coffee and tea drinkers<sup>1</sup>.

---

<sup>1</sup>Pang-Ning Tan, Michael Steinbach, and Vipin Kumar. Introduction to data mining, 2005.

# Conditional Probability, online interactive explanation

Visual Explanations Blog: conditional probability

# outline

## Conditional Probability

example 1

example 2

example 3

## Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References

# Independence, intuition

The intuition of independence is best seen mathematically via conditional probability,

$$P(B|A) = P(B).$$

That is, the probability of  $B$  does not depend on  $A$ .

# Independence

The formal definition of independence is a condition that must be checked to state that two events are independent.

## pairwise independence

Two events  $A$  and  $B$  are said to be **pairwise independent** if

$$P(A \cap B) = P(A)P(B).$$

## Independence, example

Randomly draw a card from a standard 52 card deck. Let  $A$  be the event that the card is a five and  $B$  be the event that the card is a spade. Are the events  $A$  and  $B$  independent?

$P(A \cap B) = 1/52$ . Next  $P(A) = 4/52$  and  $P(B) = 13/52$  gives  $P(A)P(B) = 1/52$ . Therefore  $P(A \cap B) = P(A)P(B)$ , so the events are independent.

# outline

Conditional Probability

example 1

example 2

example 3

Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References

## Example 1: Tulips

Suppose that we are given 20 tulip bulbs that are very similar in appearance and told that 8 tulips will bloom early, 12 will bloom late, 13 will be red, and 7 will be yellow. The following table summarizes information about the combination of features among these tulips:

|               | <i>Early</i> | <i>Late</i> | <i>Sum</i> |
|---------------|--------------|-------------|------------|
| <i>Red</i>    | 5            | 8           | 13         |
| <i>Yellow</i> | 3            | 4           | 7          |
| <i>Sum</i>    | 8            | 12          | 20         |

Q1: If one tulip bulb is selected at random, what is the probability that it will produce a red tulip?

Q2: Suppose that, under close examination, we know that our bulb will be an early bulb. What is the probability that, given we have an early bulb, it is a red tulip?



## Example 2: Conditional Calculations, Pt 1

Find  $P(A \cap B)$  if  $P(A) = 0.2$ ,  $P(B) = 0.4$ , and  $P(A|B) + P(B|A) = 0.75$ .

It may be helpful to refer to the calculations from earlier this week as reference.

## Example 3: Conditional Calculations, Pt 2

Say  $P(A \cap B) = 0.2$ ,  $P(A) = 0.6$ , and  $P(B) = 0.5$

Q1: Are A and B mutually exclusive?

Q2: Are A and B independent?

Q3: What is  $P(A^c \cap B^c)$ ?

## Example 4: Independent Calculations

Suppose that two machines 1 and 2 in a factory are operated independently of each other. Let  $A$  be the event that machine 1 will become inoperative during a given 8-hour period; let  $B$  be the event that the machine 2 will become inoperative during the same period; and suppose that  $P(A) = 1/3$  and  $P(B) = 1/4$ .

What is the probability that **at least one** of the machines will become inoperative during the given period?

Once again, knowing your formulas from earlier may be helpful here.

## Example 5: Chance Game Probability

A particular game of chance is designed so that only 20% of those that attempt it succeed. How many people need to attempt the game to have at least a 75% probability that at least one succeeds?

Hint: This problem will involve logarithms. In R, the function for natural log (log base e) is simply `log()`.

# outline

Conditional Probability

example 1

example 2

example 3

Independence

Example Problems, Pt 1

**Simulation**

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References

## Simulation Example

Assume you roll two fair six-sided dice. Let  $A$  be the event that the sum is at most 4, and  $B$  be the event that at least one die is a 2.

Find  $P(A|B) = P(A \cap B)/P(B)$  using simulation.

Recall the code we've used to perform dice rolls last week. Additionally the `"%in%"` operator may be useful here to see whether a value is in a given vector. For instance:

```
x <- 1:10  
5 %in% x
```

will give `TRUE` as a result, since 5 is a value in the vector `x`.

## Simulation Exercise

Now try a similar situation using different-sided dice. Assume you roll two fair eight-sided dice. Let  $A$  be the event that the sum is at least 12, and  $B$  be the event that at least one die is an 8.

Find  $P(B|A) = P(A \cap B)/P(A)$  using simulation.

# outline

Conditional Probability

example 1

example 2

example 3

Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References



# Law of Total Probability

The law of total probability combines this idea with the formula from conditional probability.

$$P(B) = P(A \cap B) + P(A^c \cap B)$$

Or for multiple disjoint  $A_i$  such that  $\sum_{i=1}^k P(A_i) = 1$ :

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_k \cap B).$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_k)P(B|A_k)$$

Note that all we've done is substitute each term from the first line using the fact that  $P(A \cap B) = P(A)P(B|A)$ .

## Example 1: Voting preferences

The percentage of voters classified as Liberals in three different election districts of equal size are divided as follows: 21% in the first district; 45% in the second district, and in the third district 75%. If a district is selected at random (all with equal probability of being picked) and a voter is selected at random from that district, what is the probability that they will be a Liberal?

```
liberal.given.d1 <- ____  
liberal.given.d2 <- ____  
liberal.given.d3 <- ____  
p.d1 <- ____  
p.d2 <- ____  
p.d3 <- ____  
  
(P.L <- ____)
```

## Example 2: Smoking Deaths

In a certain study it was discovered that 15% of the participants were classified as heavy smokers, 30% as light smokers, and 55% as nonsmokers. In the five year study, 20% of the heavy smokers died, 10% of the light smokers died, and 4% of the nonsmokers died. What is the probability of death for this study?

Consider which statements above make up the parts of the Law of Total Probability.

# Traffic Lights, Revisited, Pt 1

Suppose that we know, if the driver approaches a NON-green traffic light first, then the second light will be green with probability 0.10. What is the probability that a driver will find the **second** traffic light green?

# Traffic Lights, Revisited, Pt 2

We want to find the probability the second traffic light is green.

What do we know?

- ▶ Know

- ▶ probability first light is green,  $P(A) = 0.60$ .
- ▶ probability first light is not green,  $1 - P(A) = P(A^c) = 0.40$
- ▶ probability second light is green, given first light is green,  $P(B|A) = 0.80$
- ▶ probability second light is green, given first light is not green,  $P(B|A^c) = 0.10$

- ▶ Want

- ▶  $P(B)$

# outline

Conditional Probability

example 1

example 2

example 3

Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References

# Bayes' Theorem

Conditional probability combined with the law of total probability, written in all generality, is known as Bayes' Theorem.

## Bayes' Theorem

$$P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)}$$

This can also be extended to:

$$P(A_j|B) = \frac{P(A_j)P(B|A_j)}{\sum_{i=1}^k P(A_i)P(B|A_i)}$$

for any disjoint sets  $A_i$  such that  $\sum_{i=1}^k P(A_i) = 1$ .

Note that the denominator in both cases above is simply another way of writing  $P(B)$ , using the Law of Total Probability.

# outline

Conditional Probability

example 1

example 2

example 3

Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References



## Example 1: Diseases

Suppose the Department of Public Health is giving a free medical test for a certain disease. The test is 90 percent reliable in the following sense: If a person has the disease, there is a probability of .9 that the test will yield a positive response; whereas, if a person does not have the disease, there is a probability of .1 that the test will give a positive response.

Data indicate that your chances of having the disease are only 1 in 10,000. However, since the test costs you nothing, and is fast and harmless, you decide to stop and take the test. A few days later you learn that you had a positive response to the test. Now, what is the probability that you have the disease?

```
pos.given.disease <- ___  
pos.given.no.disease <- ____  
prob.disease <- ____  
prob.no.disease <- ____  
  
(prob.disease.given.positive <- ____ )
```

## Example 2: Emergency Room

At a hospital's emergency room, patients are classified and 20% of them are critical, 30% are serious, and 50% are stable.

Of the critical ones, 30% die; of the serious, 10% die; and the stable, 1% die.

Given that a patient dies, what is the conditional probability that the patient was classified as critical?

# Traffic Lights, Last Time

Find the probability that the first traffic light is green **given that** the second traffic light is green.

- ▶ Know

- ▶  $P(A) = 0.60$ .
- ▶  $P(A^c) = 0.40$
- ▶  $P(B|A) = 0.80$
- ▶  $P(B|A^c) = 0.10$
- ▶  $P(B)$  (from earlier problem)

- ▶ Want

- ▶  $P(A|B)$

# outline

Conditional Probability

example 1

example 2

example 3

Independence

Example Problems, Pt 1

Simulation

Law of Total Probability

Bayes' Theorem

Example Problems, Pt 2

References

## references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Pang-Ning Tan, Michael Steinbach, and Vipin Kumar. Introduction to data mining, 2005.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.